Fundamentals of Linear Control:
A Concise Approach

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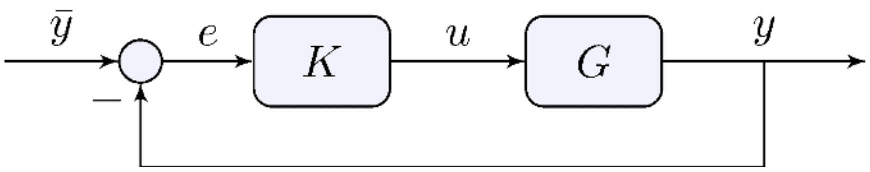
Chapter 8: Performance and Robustness

linearcontrol.info/fundamentals

8.1 Closed-Loop Stability and Performance



Tracking performance

$$e = S\bar{y}, \qquad S = \frac{1}{1+L}, \qquad L = GK$$

- Better performance needs $|S(j\omega)|$ small
- Because $H(s) + S(s) = 1$

$$S(j\omega) \approx 0 \text{ means } H(j\omega) \approx 1$$

  Beware of measurement noise
- Cannot make $S(j\omega)$ small for all ω

Tracking performance

$$e = S\bar{y}, \qquad S = \frac{1}{1+L}, \qquad L = GK$$

High gains makes S small

$$|L(j\omega)| > M \gg 1 \qquad \Longrightarrow \qquad |S(j\omega)| < \frac{1}{M-1}$$

Small gains make S one

$$|L(j\omega)| \approx 0 \qquad \Longrightarrow \qquad |S(j\omega)| \approx 1$$

e.g. L is strictly proper

$$\lim_{|s|\to\infty} |L(s)| = 0 \qquad \Longrightarrow \qquad \lim_{|s|\to\infty} |S(s)| = 1$$

Tracking performance

$$e = S\bar{y}, \qquad S = \frac{1}{1+L}, \qquad L = GK$$

What if $|L(j\omega)| \approx 1$?

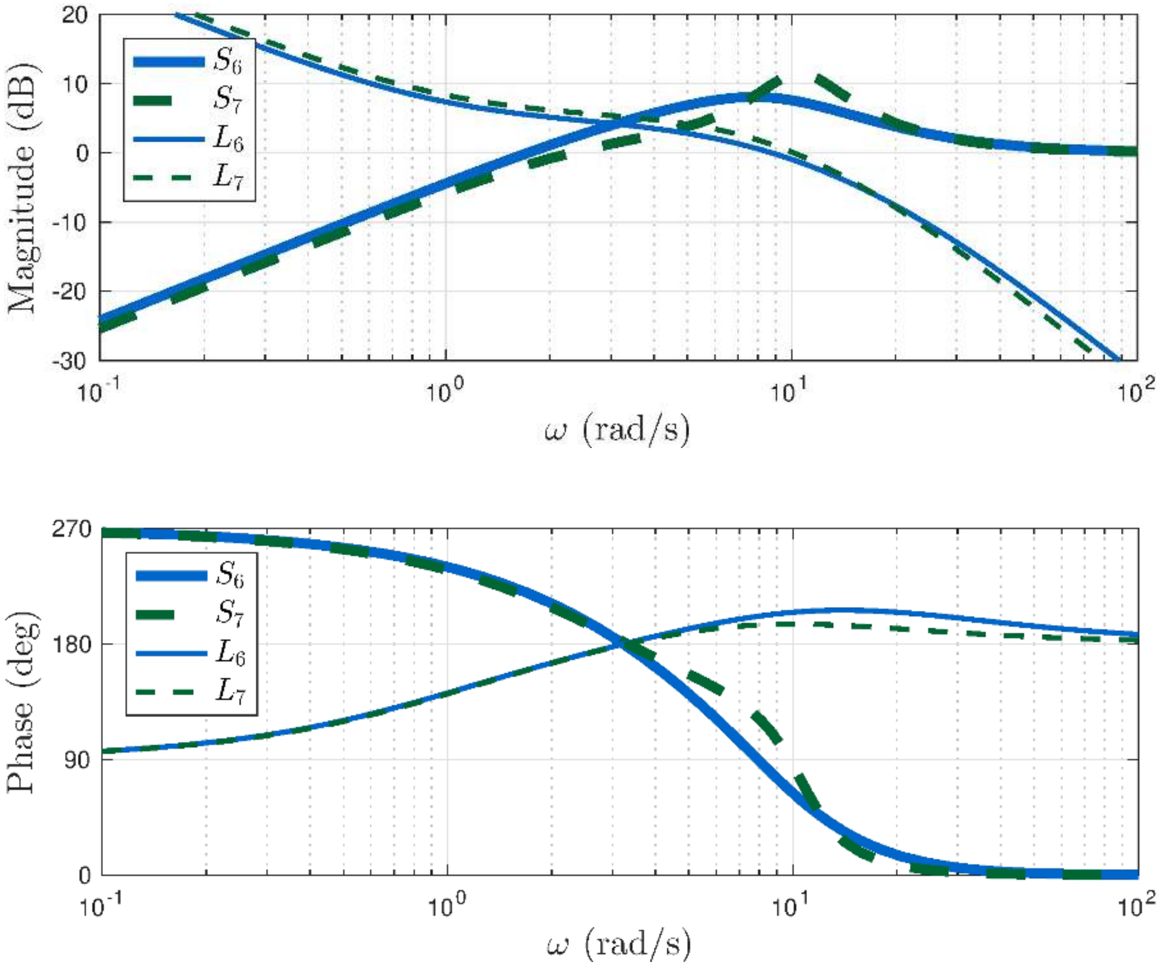
Look at the phase:

$$|L(j\omega)| \approx 1, \quad \angle L(j\omega) \approx \pi \qquad \Longrightarrow \qquad |S(j\omega)| \text{ is large}$$

S large makes S_M small

$S_M = 1/\sup_\omega |S(j\omega)|$

Example: control of the simple pendulum



What happens when $|L(j\omega)| \approx 1$?

Bode Sensitivity Integral

Theorem 8.1:

L has poles $p_i, \mathrm{Re}(p_i) > 0, i = 1, \ldots, k$. If

$$\lim_{|s|\to\infty} s\,L(s) = \kappa, \qquad |\kappa| < \infty$$

and $S = (1+L)^{-1}$ is asymptotically stable then

$$\int_0^\infty \ln|S(j\omega)|d\omega = \int_0^\infty \ln\frac{1}{|1+L(j\omega)|}d\omega = \pi\sum_{i=1}^{k} p_i - \frac{\pi\kappa}{2}$$

Main takeways:

- Closed-loop sensitivity is on a balanced budget
- Unstable systems are hard
- Choose your battles wisely!

See book for more:

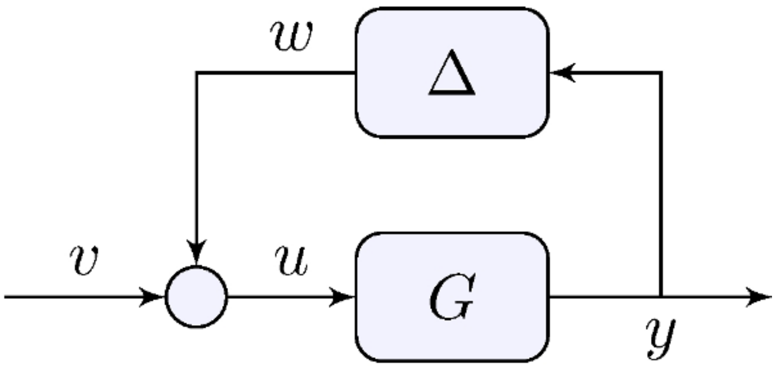
- How limited bandwidth makes things harder
- How unstable zeros amplify the effect of unstable poles

8.2 Robustness

Margins

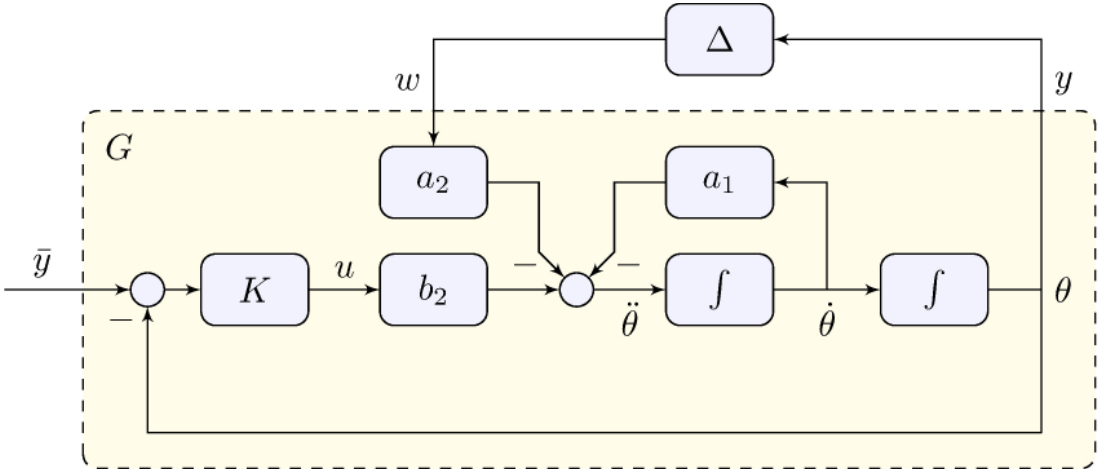
- Gain margin: robustness to changes in gain
- Phase margin: robustness to changes in phase

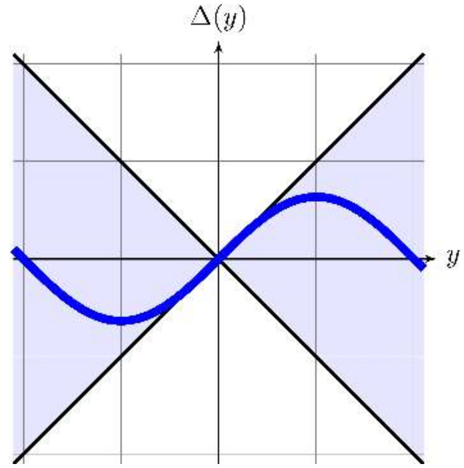
Robustness analysis



Robustness = asymptotic stability for all $\Delta \in \boldsymbol{\Delta}$

Example: control of the simple pendulum





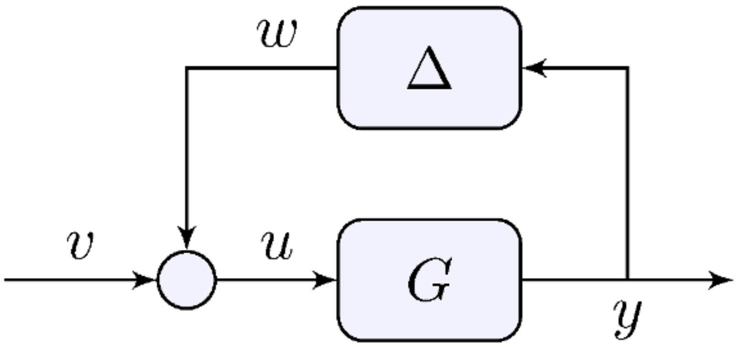
Nonlinearity as an “uncertainty”

$$\Delta(y) = \sin(y)$$

Uncertainty model

$$\boldsymbol{\Delta} = \{\Delta : \mathbb{R} \to \mathbb{R}, \quad |\Delta(y)| \le |y|\}$$

8.3 Small Gain



$$\boldsymbol{\Delta} = \{\Delta : \mathbb{R} \to \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

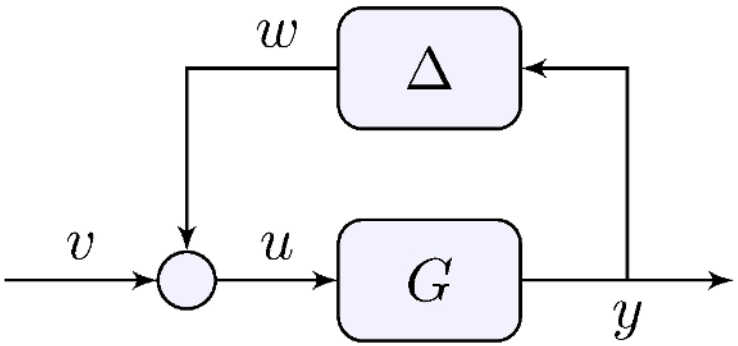
Disturbance

$$\|w\|_2 = \|\Delta(y)\|_2 \leq \|y\|_2 \quad \text{for all } \Delta \in \boldsymbol{\Delta}$$

System

$$\|y\|_2 \leq \|G\|_\infty \left(\|w\|_2 + \|v\|_2\right) + M\|x_0\|_2, \qquad \|G\|_\infty = \sup_{\omega\in\mathbb{R}} |G(j\omega)|$$

How about the closed-loop?



$$\boldsymbol{\Delta} = \{\Delta : \mathbb{R} \to \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

Small gain theorem

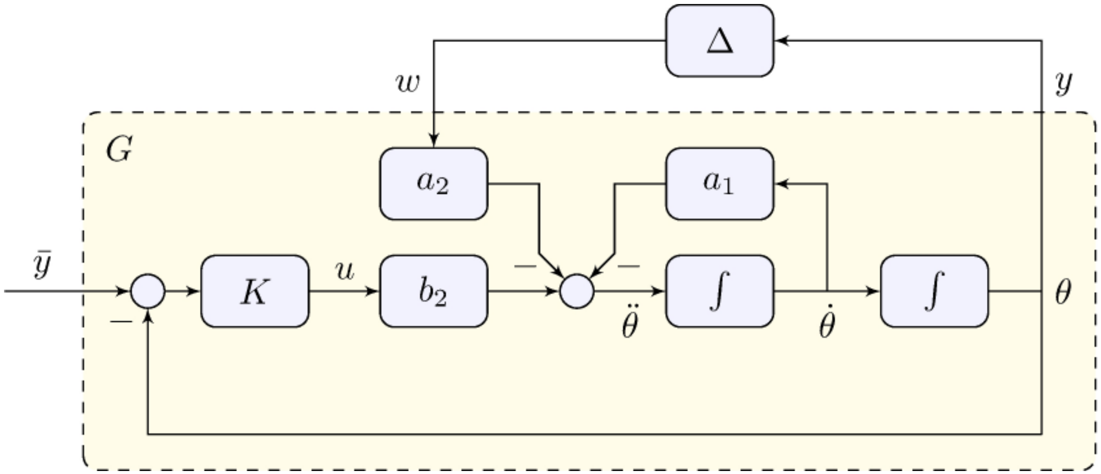
If G is asymptotically stable and $\|G\|_\infty < 1$ then

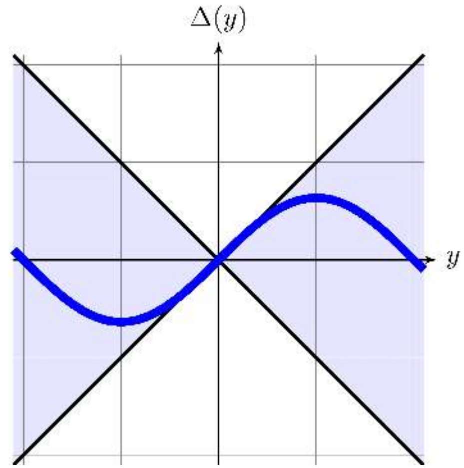
$$\|y\|_2 \leq \frac{\|G\|_\infty}{1 - \|G\|_\infty} \|v\|_2 + \frac{M}{1 - \|G\|_\infty} \|x_0\|_2$$

See book for more:

- Dynamic uncertainty
- Delays

8.4 Control of the Simple Pendulum - Part III





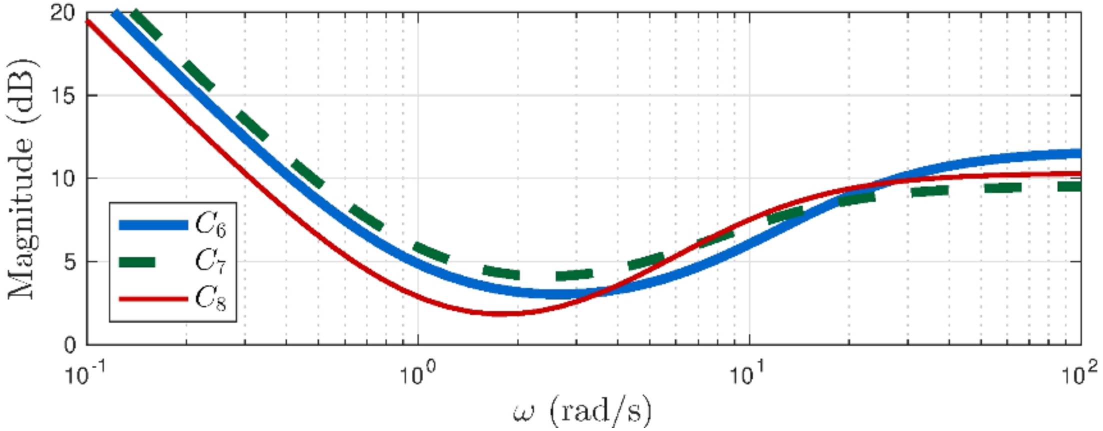
Model for robustness analysis

$$G(s) = \frac{Y(s)}{W(s)} = \frac{-a_2 F(s)}{1 + b_2 K(s) F(s)}, \qquad F(s) = \frac{1}{s^2 + a_1 s}$$

Data

$$a_1 = \frac{b}{J_r} = 0, \qquad a_2 = \frac{m\,g\,r}{J_r} \approx 49\mathrm{s}^{-2}, \qquad b_2 = \frac{1}{J_r} \approx 66.7\mathrm{kg}^{-1}\mathrm{m}^{-2}$$

Small gain analysis ($\bar{y} = 0$)

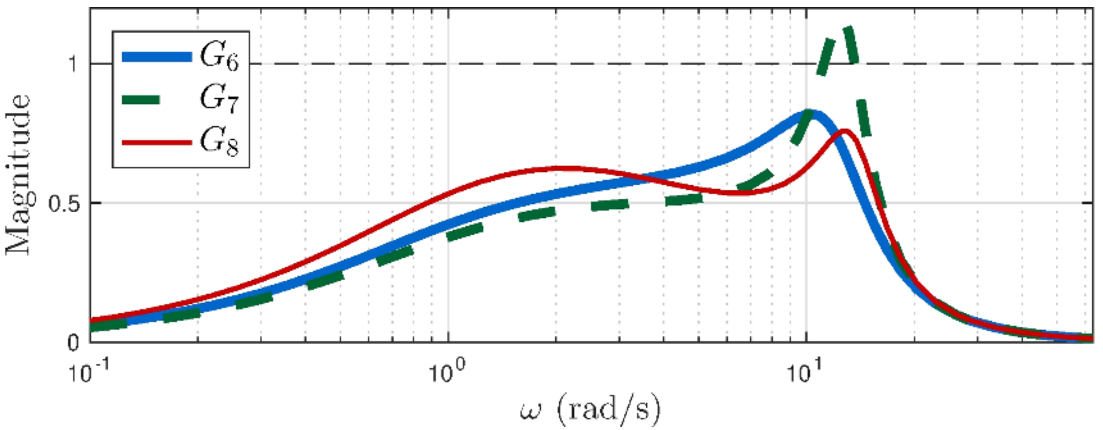


$$C_6(s) = 3.83\frac{(s+0.96)(s+6.86)}{s\,(s+21)}$$

$$C_7(s) = 3\frac{(s+1)(s+5)}{s\,(s+11)}$$

$$C_8(s) = 3.28\frac{(s+1)(s+3)}{s\,(s+10.5)}$$

- C_8: higher gains at higher frequency to lower $\|G_8\|_\infty$

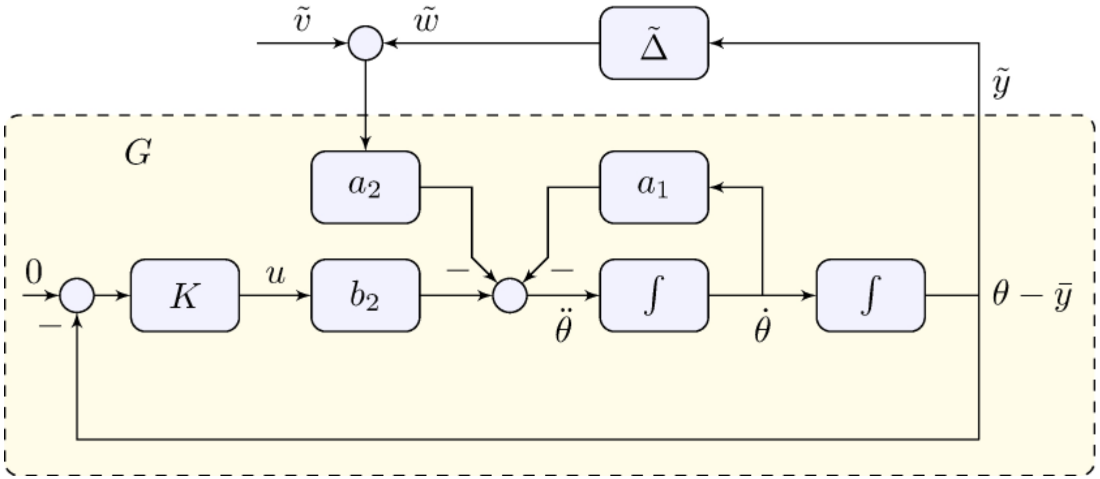


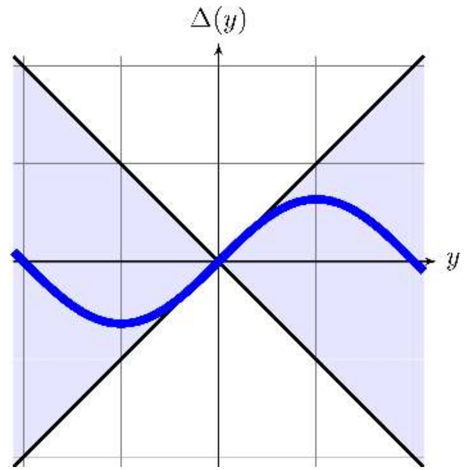
$$\|G_6\|_\infty \approx 0.82$$

$$\|G_7\|_\infty \approx 1.17$$

$$\|G_8\|_\infty \approx 0.76$$

Small gain analysis ($\bar{y} \neq 0$)





Nonlinearity

$$\tilde{\Delta}(\tilde{y}) = \cos(\bar{y})\sin(\tilde{y}) + \sin(\bar{y})(\cos(\tilde{y}) - 1), \qquad \tilde{v} = \sin(\bar{y})$$

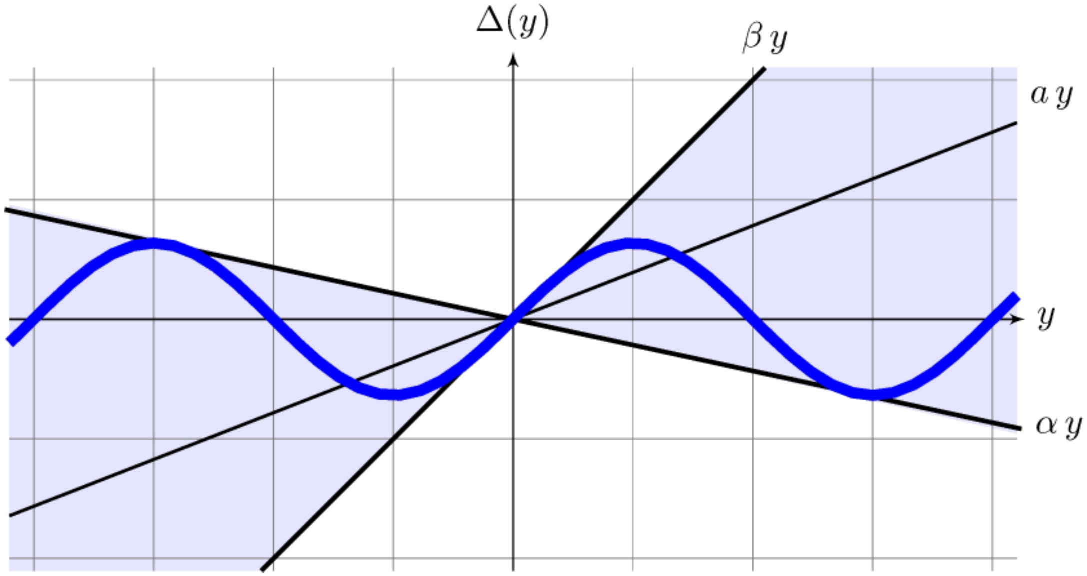
Uncertainty model

$$\tilde{\mathbf{\Delta}} = \mathbf{\Delta} = \{\Delta : \mathbb{R} \to \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

Stability analysis holds also for tracking!

See book for details and more

8.5 Circle Criterion



Sector nonlinearities

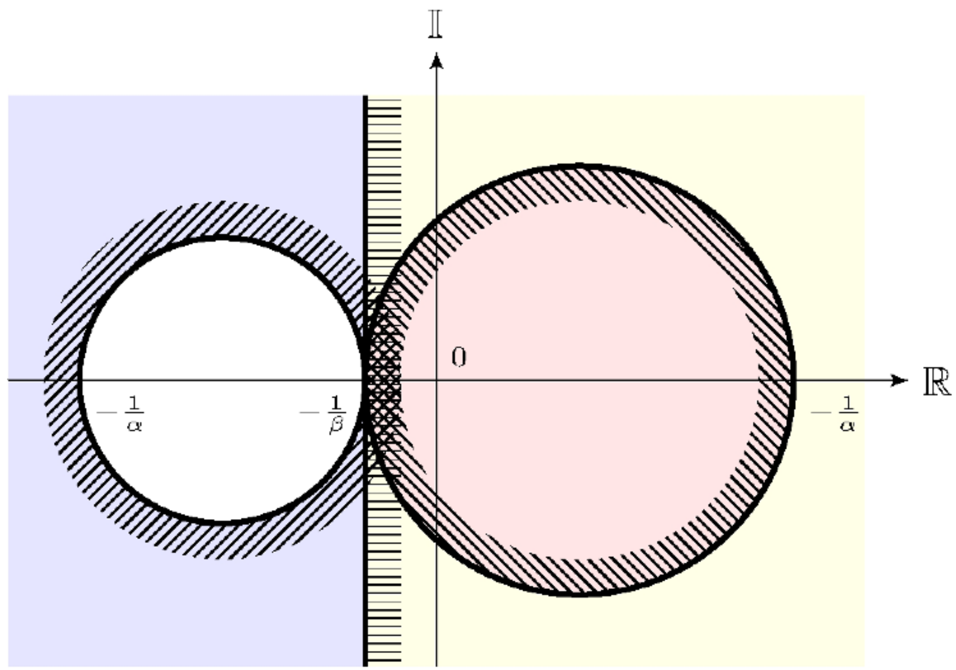
$$\mathbf{\Delta}_{ab}(a,b) = \{\Delta : \mathbb{R} \to \mathbb{R}, \quad |\Delta(y) + ay| \leq b|y|, \quad b > 0\}, \qquad (a,b) = (\tfrac{1}{2}(\alpha+\beta), \tfrac{1}{2}(\beta -$$

$$\mathbf{\Delta}_{\alpha\beta}(\alpha,\beta) = \{\Delta : \mathbb{R} \to \mathbb{R}, \quad (\Delta(y) - \alpha y)(\Delta(y) - \beta y) \leq 0, \quad \beta > \alpha\}, \qquad (\alpha,\beta) = (a-b, a+b)$$

Circle Criterion

Theorem 8.2

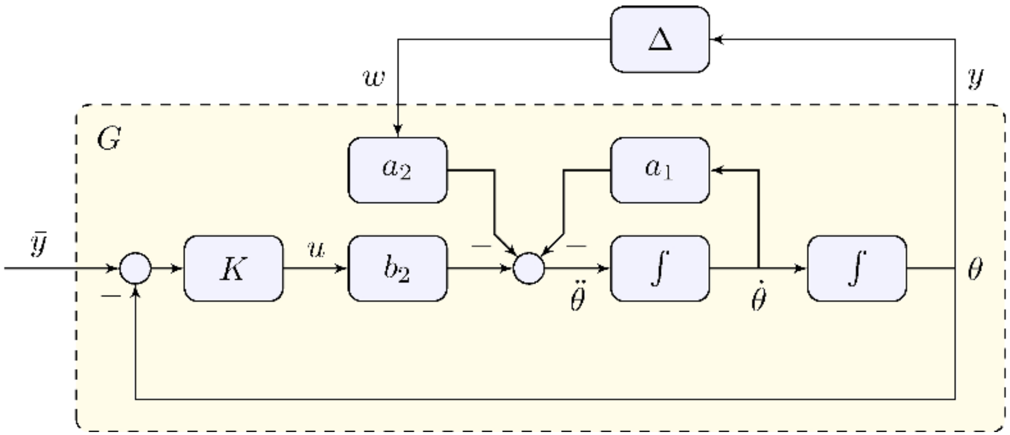
Origin is globally asymptotically stable if either

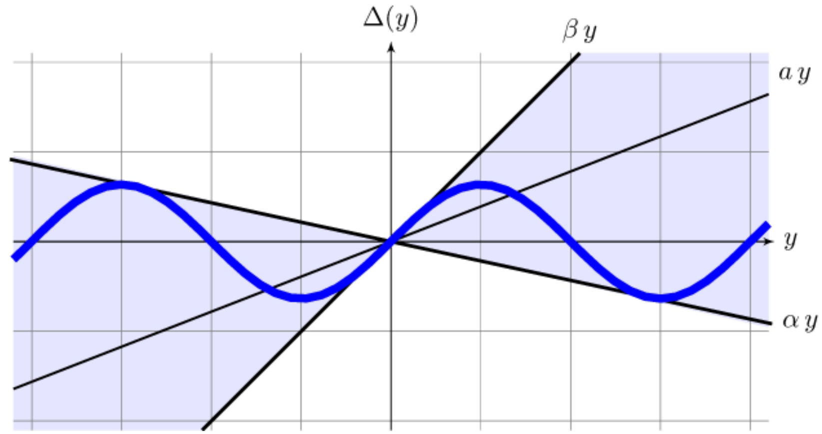


1. $\beta > 0 > \alpha$, G **is asymptotically stable and the polar plot of** G **is inside** $C(\alpha, \beta)$**;**
2. $\beta > \alpha > 0$, **the Nyquist plot of** G **encircles** $C(\alpha, \beta)$ m **times but never enters** $C(\alpha, \beta)$**.**

- m is the number of unstable poles of G

Example: control of the simple pendulum





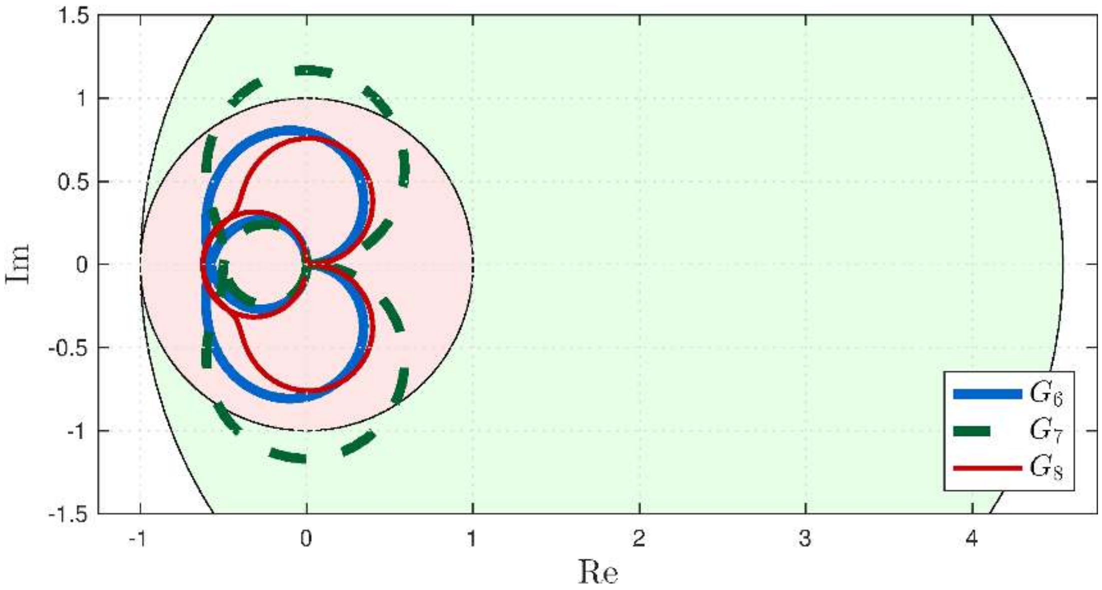
Model for robustness analysis

$$G(s) = \frac{Y(s)}{W(s)} = \frac{-a_2 F(s)}{1 + b_2 K(s) F(s)}, \qquad F(s) = \frac{1}{s^2 + a_1 s}, \qquad \alpha = -0.22, \qquad \beta = 1$$

Data

$$a_1 = \frac{b}{J_r} = 0, \qquad a_2 = \frac{m\, g\, r}{J_r} \approx 49 \mathrm{s}^{-2}, \qquad b_2 = \frac{1}{J_r} \approx 66.7 \mathrm{kg}^{-1} \mathrm{m}^{-2}$$

Example: control of the simple pendulum



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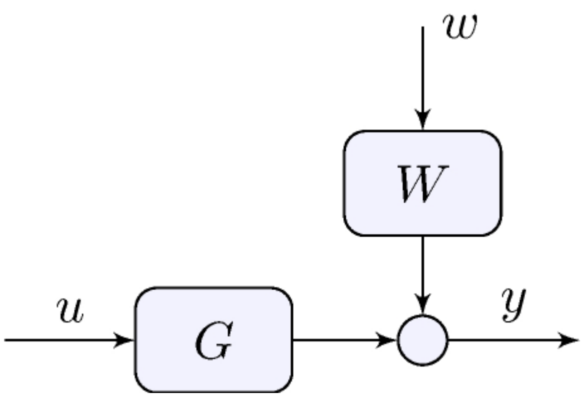
Small gain

- $C(-1, 1)$
- G_7 fails

Circle criterion

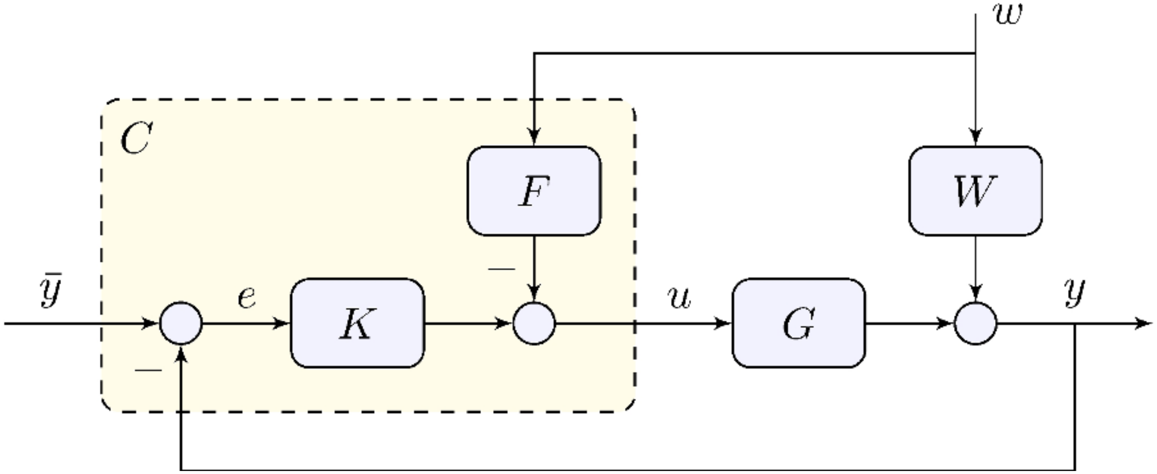
- $C(-0.22, 1)$
- All controllers stabilizing

8.6 Feedforward Control and Filtering



If the "disturbance" w is known or can be measured *feedforward* can be combined with feedback

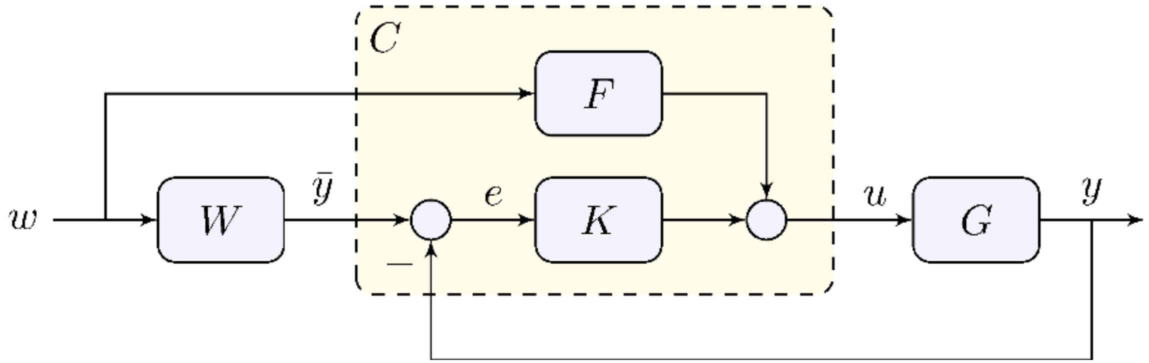
Feedforward control



Tracking error

$$e = S\,\bar{y} - (W - F\,G)\,S\,w, \qquad S = (1 + G\,K)^{-1}$$

Reference feedforward control



Tracking error

$$e = (W - F\,G)\,S\,w, \qquad S = (1 + G\,K)^{-1}$$

Feedforward control design

Beware: since F is in open-loop it has to be asymptotically stable!

If possible

$$F = G^{-1}W \qquad \Longrightarrow \qquad W = FG$$

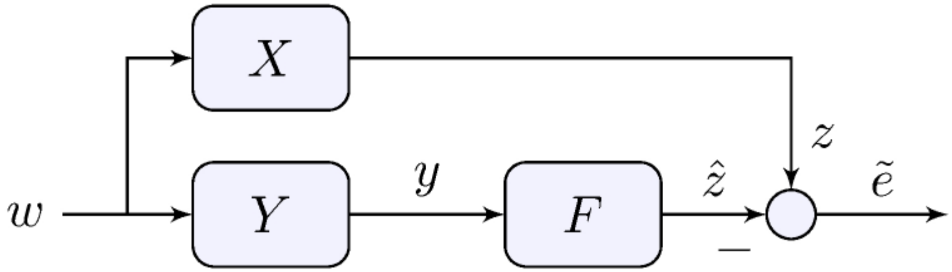
At a given frequency

$$F(j\omega) = \lim_{s \to j\omega} G(s)^{-1} W(s)$$

Optimal filter

$$\min_F \{ \|(W - FG)S\| : \quad F \text{ is asymptotically} \quad \text{stable and realizable} \}$$

Optimal filtering

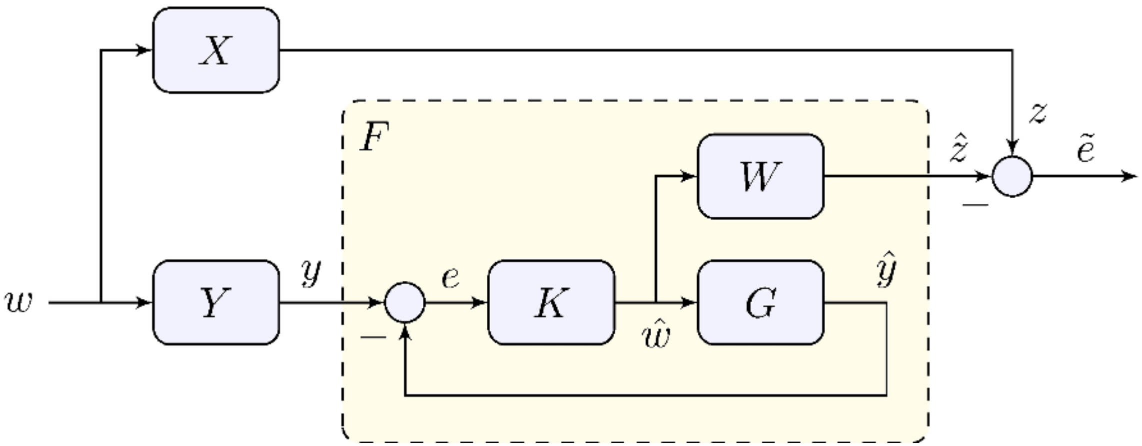


Optimization problem

$$\min_F \{ \|X - FY\| : \quad F \text{ is } \quad \text{asymptotically stable and realizable } \}$$

See book for simple examples

Optimal filtering



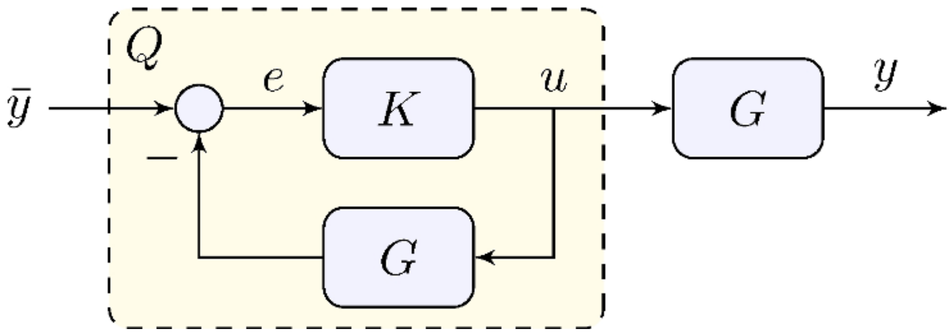
Solution via feedback

$$W = X, \qquad G = Y, \qquad \tilde{e} = W S w, \qquad S = \frac{1}{1 + G K}$$

e.g. K or G have a pole at zero then $\lim_{t \to \infty} \tilde{e}(t) = 0$

This is similar to a *Kalman filter*

All Stabilizing Controllers



Assumption: G is asymptotically stable

$$K \text{ is stabilizing} \quad \Longrightarrow \quad Q = \frac{K}{1 + K\,G} \text{ is asymptotically stable}$$

$$Q \text{ is asymptotically stable} \quad \Longrightarrow \quad K = \frac{Q}{1 - Q\,G} \text{ is stabilizing}$$

A non-trivial version exists in which G does not need to be asymptotically stable

Learn more

- Book website